

* BIG-M method * (method of penalties) Page-(90)

STEP-①: Express the problem in standard form

STEP-②: Add non-negative variable [ARTIFICIAL VARIABLE $A_1, A_2, A_3, \dots$] to the left hand side of all those constraints which are of $\geq$ or $=$ type.

STEP-③: Assign a very large PENALTY ($-M$) to these artificial variables in the objective function.

STEP-④: solve the modified LPP by SIMPLEX METHOD

NOTE: Once an artificial variable leaves the basis, the column for this variable is **OMITTED** from the next simplex table

At any iteration of simplex method, one of the following three cases may arise

Sl. No.	POSSIBILITIES	CONCLUSION
1	NO ARTIFICIAL VARIABLE in the BASIS + OPTIMALITY CONDITION is satisfied	Solution is OPTIMAL BASIC FEASIBLE SOLUTION
2	There is at least ONE artificial variable in BASIS at ZERO LEVEL with zero value in b column + OPTIMALITY CONDITION is satisfied	Solution is DEGENERATE OPTIMAL BASIC FEASIBLE SOLUTION
3	There is at least ONE artificial variable in BASIS at NON ZERO LEVEL with positive value in b column + OPTIMALITY CONDITION is satisfied	Solution is NO FEASIBLE SOLUTION

Q) solve the following LPP by BIG-M method

$$\text{minimize } Z = 2x_1 + x_2$$

$$\text{subject to: } 3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 3$$

$$x_1, x_2 \geq 0$$

First convert it to MAXIMIZATION problem

$$\text{max } Z' = -2x_1 - x_2$$

Here Z' is $(-Z)$

Soln:- consider the first constraint:

$$3x_1 + x_2 = 3$$

(= Type)

∴ We have to add
ARTIFICIAL VARIABLE

$$3x_1 + x_2 + A_1 = 3$$

consider the second constraint:

$$4x_1 + 3x_2 \geq 6$$

(≥ Type)

Introduce SURPLUS VARIABLE S_1

+

Add Artificial variable A_2

∴

$$4x_1 + 3x_2 - S_1 + A_2 = 6$$

consider the third constraint:

$$x_1 + 2x_2 \leq 3$$

We have to just introduce
SLACK VARIABLE s_2 here

$$\therefore x_1 + 2x_2 + s_2 = 3$$

Thus, modified LPP is

$$\text{MAX } Z' = -2x_1 - x_2 + 0s_1 + 0s_2 - MA_1 - MA_2$$

$$\begin{aligned} \text{subject to: } & 3x_1 + x_2 + 0s_1 + 0s_2 + A_1 + 0A_2 = 3 \\ & 4x_1 + 3x_2 - s_1 + 0s_2 + 0A_1 + A_2 = 6 \\ & x_1 + 2x_2 + 0s_1 + s_2 + 0A_1 + 0A_2 = 3 \end{aligned}$$

$$x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$$

INITIAL SIMPLEX TABLE:

Page-(25)

Surplus variable S_1 is not a basic variable.

Because, when initially $x_1=0$ & $x_2=0$ (Non-basic),

from second constraint, we get $S_1 = -6$.

As negative quantities are not feasible, S_1 must be prevented from appearing in the initial solution.

This is done by taking $S_1 = 0$

BASIC
VARIABLE

∴ Initial basic feasible solution is:

$$x_1=0, x_2=0, S_1=0$$

Non-basic variable

$$A_1=3, A_2=6, S_2=3$$

↓
from 1st
condition

↓
from 2nd
condition

↓
from 3rd
condition

∴ The initial simplex table is:

C_B	C_j BASIS	-2 x_1	-1 x_2	0 s_1	0 s_2	-M A_1	-M A_2	b	θ
-M	A_1	3	1	0	0	1	0	3	$3/3 = 1$
-M	A_2	4	3	-1	0	0	1	6	$6/4 = 1.5$
0	s_2	1	2	0	1	0	0	3	$3/1 = 3$
$Z_j = \sum C_B x_j$		-7M	-4M	M	0	-M	-M	-9M	
$C_j - Z_j$		7M-2	4M-1	-M	0	0	0		

← MINIMUM POSITIVE RATIO

∴ outgoing variable is

A_1

Delete A_1 column in the next table

↑ LARGEST POSITIVE VALUE (∵ M is LARGE)
∴ Incoming variable is x_1

Iteration (01):

Page-(27)

Introduce x_1 & drop A_1 from the BASIS.

Also, Divide ROW(1) by the key element 3

C_B	C_j BASIS	-2 x_1	-1 x_2	0 s_1	0 s_2	-M A_2	b	θ
-2	x_1	1	$\frac{1}{3}$	0	0	0	1	
-M	A_2							
0	s_2							
Z_j								
$C_j - Z_j$								

Now convert all the elements of **KEY COLUMN**

Page-(28)

to zero using elementary row operation.

$$R_2 \rightarrow R_2 - 4(\text{NEW } R_1)$$

$$R_3 \rightarrow R_3 - (\text{NEW } R_1)$$

C_B	C_j BASIS	-2 x_1	-1 x_2	0 s_1	0 s_2	-M A_2	b	θ
-2	x_1	1	1/3	0	0	0	1	$1/(1/3) = 3$
-M	A_2	0	5/3	-1	0	1	2	$\frac{2}{(5/3)} = \frac{6}{5}$
0	s_2	0	5/3	0	1	0	2	$\frac{2}{(5/3)} = \frac{6}{5}$
Z_j		-2	$-\frac{2}{3} + \frac{5M}{3}$	M	0	-M	-2-2M	
$C_j - Z_j$		0	$\frac{5M}{3} - \frac{1}{3}$	-M	0	0		

↑ LARGEST +ve value
∴ INCOMING VARIABLE is x_2

∴ outgoing variable is A_2
delete A_2 column from next table

← minimum POSITIVE RATIO

Iteration - (02):

page - (29)

Introduce x_2 and drop A_2 from the basis

Also, divide ROW - (2) by the key element $5/3$

C_B	C_j BASIS	-2 x_1	-1 x_2	0 s_1	0 s_2	b	θ
-2	x_1						
-1	x_2	0	1	$-\frac{3}{5}$	0	$\frac{6}{5}$	
0	s_2						

NOW, convert all the elements of **KEY COLUMN** to ZERO using elementary row operation

$$R_1 \rightarrow R_1 - \left(\frac{1}{3}\right) \text{NEW } R_2, \quad R_3 \rightarrow R_3 - \left(\frac{5}{3}\right) \text{NEW } R_2$$

C_B	C_j BASIS	-2 x_1	-1 x_2	0 s_1	0 s_2	b	θ
-2	x_1	1	0	1/5	0	3/5	
-1	x_2	0	1	-3/5	0	6/5	
0	s_2	0	0	1	1	0	
Z_j		-2	-1	1/5	0	$-\frac{12}{5}$	
$C_j - Z_j$		0	0	-1/5	0		

Observe that none of $C_j - Z_j$ value
is positive

Page - (31)

∴ This is OPTIMAL SOLUTION

From the BASIS column & b column, we have

BASIS	b
x_1	$3/5$
x_2	$6/5$
s_2	0
Z_j	$-12/5$

∴ $x_1 = \frac{3}{5}$, $x_2 = \frac{6}{5}$

$\max Z' = -\frac{12}{5}$

But, the given problem is
of minimization type

∴ $\min(Z) = -\max(Z') = \frac{12}{5}$